

ZIXIN GUO

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EDUCATION

Carnegie Mellon University

Master of Science in Robotics

Aug. 2025 – Present

Pittsburgh, PA

- Advisor: Prof. Deva K. Ramanan
- GPA: 4.20/4.00
- **Relevant Coursework:** Mobile Robotics, Robot Learning

University of Toronto

Honours Bachelor of Science

Sept. 2020 – Jun. 2025

Toronto, ON

- **Program:** Computer Science Specialist
- **GPA:** 3.98/4.0 (91.5%)

TECHNICAL SKILLS

Vision & Generative Models: Video Diffusion, Generative Modeling, 3D Reconstruction, NeRF, 3D Gaussian Splatting
Robot Learning: Reinforcement Learning, Motion Retargeting, Trajectory Tracking, Simulation-based Evaluation
Frameworks & Tools: PyTorch, CUDA, Diffusers, OpenCV, MuJoCo, Isaac Lab, Git, Slurm, Vim, React, Django
Programming: Python, C/C++, Java, Shell, MySQL, JavaScript, HTML, CSS, LaTeX, R

PUBLICATIONS

Z. Guo, Y. Litman, Y. He, J. Miller, C. Chen, D. Ramanan, “LightCrafter: PBR-Conditioned Video Diffusion Refinement for Controllable and Consistent Relighting,” submitted to *Conference on Neural Information Processing Systems (NeurIPS)*, 2026, arXiv: 2607.08016.

Z. Guo, J. Zhang, “FastTalker: Jointly Generating Speech and Conversational Gestures from Text,” in *European Conference on Computer Vision Workshop (ECCVW)*, 2024, arXiv: 2409.16404.

D. Huo, **Z. Guo**, X. Zuo, Z. Shi, J. Lu, P. Dai, S. Xu, L. Cheng, Y.-H. Yang, “TexGen: Text-Guided 3D Texture Generation with Multi-view Sampling and Resampling,” in *European Conference on Computer Vision (ECCV)*, 2024, arXiv: 2408.01291.

Z. Guo, R. Yang, “A Channel Attention and Feature Manipulation Network for Facial Expression Recognition,” in *3rd International Conference on Signal Processing and Machine Learning*, 2023 (Oral Presentation), DOI: 10.54254/2755-2721/6/20230751.

EXPERIENCES

Center for Autonomous Vehicle Research, Carnegie Mellon University

Graduate Student Researcher. Supervisor: Prof. Deva K. Ramanan

Sept. 2025 - Present

Pittsburgh, PA

- Proposed LightCrafter, a hybrid video relighting pipeline that reformulates the task as translation of a physically-based rendering (PBR) proxy, baking the target illumination into the proxy for finer lighting control and long-form temporal consistency.
- Post-trained CogVideoX on synthetic video pairs and real-world unpaired videos to recover hard-to-model effects such as global illumination that PBR renders alone fail to reproduce.
- Outperformed prior state-of-the-art methods on real-world relighting benchmarks and contributed a synthetic benchmark, evaluation metrics, and dataset for further analysis.
- Developing a video-conditioned model that predicts G1 humanoid reference trajectories directly from monocular human demonstrations.
- Evaluating predicted trajectories through closed-loop simulation with SONIC, comparing motion-tracking performance against a two-stage GVHMR-to-GMR baseline.

Machine Learning and Computational Healthcare Lab, Vector Institute

Undergraduate Student Researcher. Supervisor: Prof. Rahul G. Krishnan

Sept. 2024 - Jan. 2025

Toronto, ON

- Conducted a literature review on various model architectures, spanning from early designs like LSTM, through transformers, to large language models (LLMs), in the context of medical time series.
- Developed a codebase and conducted benchmark studies on various time series forecasting models and data imputation methods, applying them to datasets such as MIMIC and HiRID.
- Demonstrated explainability in patient vital sign predictions and experimented with the performance impact of data imputation using LLM as a zero-shot predictor.

Dynamic Graphics Project Lab, University of Toronto

UTEA Undergraduate Student Researcher. Supervisor: Prof. David Lindell

May 2024 - Nov. 2024

Toronto, ON

- Proposed a novel text-to-image diffusion-based optimization method that learns age-aware component-level concepts given a collection of person-specific images spanning years.
- Performed comprehensive quantitative analysis and user studies to evaluate multiple applications of the proposed method, such as recontextualization, component-level swapping, age regression, and progression. Demonstrated enhanced age-specific component control in the personalization process over baseline methods.

- Work resulted in a first-author poster presentation at Applied Research in Action (ARIA) 2024.

Noah's Ark Lab

May 2023 - May 2024

Computer Vision Research Intern. Supervisor: Dr. Juwei Lu

Toronto, ON

- Developed and implemented a 3D scene editing pipeline using 3D Gaussian Splatting (3DGS) and diffusion models for smart factory modeling as an alternative to the LiDAR-based method. This proof of concept was successfully transitioned to the production department.
- Proposed and initiated a follow-up project on 3D object removal and view-consistent inpainting in 3DGS, which aligns with both research and industry domains. The work resulted in a US patent submission.
- Contributed to a research project on diffusion-based 3D mesh texturing by developing the Attention-Guided Multi-View Sampling module, leading all baseline benchmark studies, and drafting the initial paper. The work resulted in an ECCV 2024 conference paper.

Dynamic Graphics Project Lab, University of Toronto

Jan. 2023 - May 2023

Undergraduate Research Assistant. Supervisor: Prof. David Lindell

Toronto, ON

- Developed and replicated a diffusion-based model from scratch for 3D-aware neural field generation, evaluating various 3D neural representations, such as SDFs, beyond the occupancy fields outlined in the original work.
- Evaluated the model's performance by benchmarking it against Chamfer-L1 and Volumetric IoU metrics to analyze the surface geometry and volumetric alignment of the generated 3D shapes.
- Work ended with a technical report: Evaluating 3D Neural Field Diffusion Model Using Triplanes and served as a baseline for other research projects in the group.

CV Lab, JD.com

May 2021 - Sept. 2021

Computer Vision Research Intern. Supervisor: Dr. Tao Mei

Beijing, China

- Proposed a novel self-attention defect detection method integrating a Vision Transformer (ViT) with a memory module for industrial quality inspection.
- Conducted detailed experiments using the proposed method that achieved state-of-the-art results on the SDNET 2018 dataset using just 50% of the training data. This work subsequently led to a published patent.
- Competed in the CVPR 2021 workshop challenge on open-set image classification; identified optimal settings across dozens of combinations to enhance classification accuracy, securing 3rd place on the leaderboard.

PROJECTS

Object Editing with 3D Scene Representation and Rendering

Jan. 2024 - Apr. 2024

- Adopted the Segment Anything (SAM) on a single reference view for editing region segmentation, propagated the segmentation mask to remaining views via depth warping, and performed 3D semantic mask reconstruction to accurately localize edited regions within the 3D scene.
- Proposed utilizing 2D editing techniques and attention-guided control to achieve smooth and view-consistent 3D editing. The edited images are then utilized to fine-tune the 3D scene, represented by the neural radiance field (NeRF) or 3DGS.

"An Image is Worth One Sentence": Fast Textual Inversion with Supreme Initialization

Jan. 2023 - Apr. 2023

- Improved textual inversion, a state-of-the-art image personalization method, by increasing its convergence speed from 5000 steps to 100 steps.
- Engineered the Image Personalization feature, dynamically customizing images to align with the user's prompt.
- Pioneered the multitoken initialization method and the class/caption initialization method.

Smart Oilfield

May 2022 - Aug. 2022

- Engaged oil and gas experts to acquire professional knowledge and annotated a large set of surveillance videos captured at the PetroChina Oilfield for anomaly detection.
- Fine-tuned YOLO series detection models on the annotated datasets for abnormal activity recognition at the construction site, such as PPE wearing, smoke detection, and illegal trespassing.
- Demoed and deployed the trained models on the Atlas 500 AI Edge Station and achieved a 30 FPS detection rate on the monitoring platform.

AWARDS AND HONOURS

Graduate Research Assistantship, Robotics Institute, Carnegie Mellon University (Awarded \$42,167 USD)	2025 - 2026
University of Toronto Excellence Awards (UTEA) (Awarded \$7500 CAD)	2024
Dean's List Scholar at the University of Toronto	2021, 2022, 2023, 2024
Star Intern Certificate for Self-Attention Defect Detection by the VP Dr. Tao Mei of JD.com	2021
3rd Place in Open-World Image Classification Challenge at CVPR 2021 Workshop	2021
President's Scholar of Excellence Program at the University of Toronto (Awarded \$10000 CAD)	2020
Top 2% in the Senior Level Canadian Computing Competition	2020
Top 3% in the Euclid Contest and Canadian Senior Math Competition	2020